Shadow Governance: Coercion-Proof Voting Under Adversarial Surveillance

Encrypted Ballots with Delayed Revelation for the Pars Network

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Abstract

We present ShadowVote, a coercion-resistant electronic voting protocol for the Pars Network that enables diaspora community members to participate in governance even under adversarial surveillance and coercion. ShadowVote provides four properties simultaneously: (1) *ballot secrecy*—no one learns an individual's vote, (2) *receipt-freeness*—voters cannot prove how they voted to a coercer, (3) *coercion resistance*—voters can submit a vote that appears to follow coercer instructions while actually recording their true preference, and (4) *universal verifiability*—anyone can verify the election result is correct. Our protocol uses a novel combination of ElGamal re-encryption with time-locked commitments and deniable encryption, enabling a two-phase voting process where encrypted ballots are submitted during the voting period and revealed after a delay. We formalize the security model for diaspora-specific coercion scenarios (device seizure, forced delegation, familial pressure) and prove that ShadowVote satisfies all four properties under the DDH assumption. Evaluation with 3,200 simulated voters demonstrates negligible computational overhead (380ms per vote on mobile devices) and resistance to adaptive coercion strategies.

1 Introduction

Electronic voting under coercion is a foundational problem in cryptography [Juels et al., 2005, Benaloh and Tuinstra, 1994]. For diaspora communities, the problem takes on urgent practical dimensions: members in authoritarian jurisdictions may face state pressure to vote in specific ways, delegate their governance power to regime-aligned actors, or abstain entirely from community governance [Dehghan, 2018]. Even in democratic host countries, social and familial pressure within tight-knit diaspora communities can compromise the free exercise of governance rights.

Standard e-voting protocols provide ballot secrecy but typically fail at receipt-freeness and coercion resistance. A coerced voter can be forced to reveal their randomness, share their screen during voting, or use a specific device under the coercer's control. These attack vectors are not theoretical—they are documented practices in authoritarian contexts [Hermet et al., 1978].

1.1 Diaspora-Specific Coercion Scenarios

We identify six coercion scenarios specific to the Persian diaspora:

S1: **State surveillance:** A government monitors network traffic of citizens abroad to detect participation in diaspora governance.

S2: **Forced delegation:** A regime agent forces a user to delegate their veASHA voting power to a controlled account.

S3: **Device seizure:** At a border crossing, authorities seize a device and demand the user demonstrate how they voted.

S4: **Familial pressure:** Family members in Iran pressure relatives abroad to vote in alignment with regime preferences to protect family at home.

S5: **Vote buying:** An actor offers financial incentives for verifiable vote commitments.

S6: **Retrospective punishment:** After a vote, authorities identify voters who opposed regime interests and retaliate against their families.

1.2 Contributions

(i) A formal coercion model capturing the six diaspora-specific scenarios.

(ii) The ShadowVote protocol providing ballot secrecy, receipt-freeness, coercion resistance, and universal verifiability.

(iii) A novel *deniable ballot* construction allowing voters to produce fake credentials that decrypt to any chosen vote.

(iv) Time-locked commitment schemes enabling delayed revelation that bounds the coercer's observation window.

(v) Security proofs under the DDH assumption with simulation-based definitions.

(vi) Practical evaluation demonstrating feasibility on mobile devices.

2 Background

2.1 Coercion-Resistant Voting

Juels, Catalano, and Jakobsson (JCJ) [Juels et al., 2005] introduced the first formal treatment of coercion-resistant voting. Their protocol provides a *fake credential* that a voter can give to a coercer; votes cast with fake credentials are silently discarded during tallying. Civitas [Clarkson et al., 2008] extended JCJ with improved credential management. Selene [Ryan et al., 2016] provides a verifiable receipt that the voter can use to check their vote but that is useless to a coercer.

2.2 Receipt-Free Voting

Benaloh and Tuinstra [Benaloh and Tuinstra, 1994] formalized receipt-freeness: no voter should be able to construct a receipt proving their vote. Their construction uses an untappable channel between voter and authority. Hirt and Sako [Hirt and Sako, 2000] proposed receipt-free schemes based on homomorphic encryption. Moran and Naor [Moran and Naor, 2006] achieved receipt-freeness without untappable channels using physical assumptions.

2.3 Time-Locked Cryptography

Time-lock puzzles [Rivest et al., 1996] force a minimum computational delay before a secret can be recovered. Verifiable delay functions (VDFs) [Boneh et al., 2018] provide publicly verifiable time-locked outputs. Timed commitments [Boneh and Naor, 2000] combine commitments with time-lock properties.

2.4 Deniable Encryption

Deniable encryption [Canetti et al., 1997] allows a sender to produce a ciphertext that can be "opened" to multiple plaintexts, each with its own plausible randomness. Under coercion, the sender reveals fake randomness that decrypts to a coercer-approved message while the true message remains hidden.

3 System Model

3.1 Participants

Definition 1 (System Participants).

- ***Voters:*** *$\mathcal{V} = \{v_1, \ldots, v_n\}$ with veASHA-weighted voting power $\{w_1, \ldots, w_n\}$.*
- ***Tallying Committee:*** *$\mathcal{T} = \{t_1, \ldots, t_m\}$ ($m = 7$) jointly hold the election decryption key via $(4, 7)$-threshold sharing.*
- ***Bulletin Board:*** *A public, append-only ledger (the Pars blockchain) where all encrypted ballots and proofs are posted.*

- ***Coercer:*** *$\mathcal{C}$ who can observe the voter's actions, demand credential disclosure, and verify claimed vote behavior.*

3.2 Formal Coercion Model

Definition 2 (Coercion Capability)**.** *The coercer $\mathcal{C}$ has the following powers:*

(a) ***Observation:*** *$\mathcal{C}$ can observe the voter's screen and network traffic during voting.*

(b) ***Credential demand:*** *$\mathcal{C}$ can demand the voter reveal their secret key material after voting.*

(c) ***Vote instruction:*** *$\mathcal{C}$ can instruct the voter to vote for a specific option.*

(d) ***Forced abstention:*** *$\mathcal{C}$ can attempt to prevent the voter from voting.*

(e) ***Retrospective:*** *$\mathcal{C}$ can analyze the public bulletin board after the election to correlate votes with voters.*

$\mathcal{C}$ cannot physically prevent the voter from accessing the voting system through an alternative channel (e.g., a different device or time).

3.3 Security Definitions

Definition 3 (Coercion Resistance (Simulation-Based))**.** *A voting protocol Π is coercion-resistant if there exists a simulator $\mathcal{S}$ such that for any PPT coercer $\mathcal{C}$, the output of $\mathcal{S}$ interacting with $\mathcal{C}$ is computationally indistinguishable from the output of the real protocol:*

$$\mathsf{REAL}_{\Pi,\mathcal{C}}(\lambda) \approx_c \mathsf{IDEAL}_{\mathcal{S},\mathcal{C}}(\lambda) \tag{1}$$

where λ is the security parameter. The simulator can cause the voter to appear to follow the coercer's instructions while the tallied vote reflects the voter's true preference.

Definition 4 (Receipt-Freeness)**.** *A protocol is receipt-free if for any two votes v_0, v_1 and any PPT distinguisher $\mathcal{D}$:*

$$|\mathbb{P}[\mathcal{D}(\mathsf{view}_{\mathcal{C}}(\mathsf{Vote}(v_0))) = 1] - \mathbb{P}[\mathcal{D}(\mathsf{view}_{\mathcal{C}}(\mathsf{Vote}(v_1))) = 1]| \leq \mathsf{negl}(\lambda) \tag{2}$$

where $\mathsf{view}_{\mathcal{C}}$ is the coercer's view of the voting process.

Definition 5 (Universal Verifiability)**.** *A protocol is universally verifiable if any observer with access to the bulletin board can verify that:*

(i) Only eligible voters cast ballots.

(ii) Each voter cast at most one ballot.

(iii) The tally correctly reflects the cast ballots.

4 The Shadow Vote Protocol

4.1 Setup Phase

Algorithm 1 Election Setup

Require: Security parameter λ, voter registry $\mathcal{V}$, tallying committee $\mathcal{T}$
Ensure: Election parameters published on bulletin board

```
Generate ElGamal parameters: (𝔾, g, q) ← ParamGen(1^λ)
Tallying committee generates threshold key:
    Each t_j generates (sk_j, pk_j) ← KeyGen(𝔾)
    Combined public key: pk_E = ∏_j pk_j
Generate time-lock parameters:
    T_reveal: reveal deadline (voting end + 48 hours)
    Δ: VDF difficulty (calibrated to 48 hours on standard hardware)
Publish (pk_E, T_reveal, Δ, 𝒱, options) to bulletin board
Each voter v_i receives credential: cred_i ← IssueCredential(v_i, pk_E)
```

4.2 Credential System

Each voter receives a credential that can be "cloned" to produce a fake credential for the coercer:

Definition 6 (Deniable Credential)**.** *A credential for voter v_i is a tuple $cred_i = (x_i, Y_i, \sigma_i)$ where:*

- *$x_i \in_R \mathbb{Z}_q$ is the secret credential key.*
- *$Y_i = g^{x_i}$ is the public credential.*
- *σ_i is the registration authority's BBS+ signature on $(Y_i, v_i.did, w_i)$.*

A fake credential *$cred'_i = (x'_i, Y'_i, \sigma'_i)$ is constructed such that:*

- *$x'_i \neq x_i$ but σ'_i appears valid to the coercer.*
- *Votes cast with $cred'_i$ are detected and discarded during tallying.*
- *Given $cred'_i$, the coercer cannot distinguish it from $cred_i$.*

Algorithm 2 Fake Credential Generation

Require: Real credential $cred_i = (x_i, Y_i, \sigma_i)$
Ensure: Fake credential $cred'_i$ indistinguishable from real

```
x'_i ←_R ℤ_q                          ▷ Random fake secret
Y'_i ← g^{x'_i}                       ▷ Fake public credential
σ'_i ← SimulateBBS(Y'_i)              ▷ Simulated signature
return cred'_i = (x'_i, Y'_i, σ'_i)
```

The key insight: the simulated BBS+ signature σ'_i is indistinguishable from a real signature under the q-SDH assumption, but it is not in the valid credential registry. During tallying, votes with unregistered credentials are silently removed.

4.3 Vote Casting

Algorithm 3 Shadow Vote Casting

Require: Voter v_i, true vote $v \in \{1, \ldots, k\}$, credential $cred_i$
Ensure: Encrypted ballot posted to bulletin board

Step 1: Encode vote as group element
$M_v \leftarrow g_v$ where g_v is the generator for option v
Step 2: Encrypt vote
$r \leftarrow_R \mathbb{Z}_q$
$(c_1, c_2) \leftarrow (g^r, M_v \cdot pk_E^r)$ ▷ ElGamal encryption
Step 3: Create time-locked commitment
$puzzle \leftarrow \mathsf{TimeLock}(r, \Delta)$ ▷ Time-lock the randomness
$com \leftarrow \mathcal{H}(v \| r \| x_i)$ ▷ Binding commitment
Step 4: Generate validity proof
$\pi_1 \leftarrow \mathsf{ZKProof}\{(v, r, x_i) : c_2 = g_v \cdot pk_E^r \wedge Y_i = g^{x_i}\}$ ▷ Prove: vote is valid and credential is known
Step 5: Create deniable layer
$v' \leftarrow$ coercer-instructed vote (or random if no coercion)
$r' \leftarrow \mathsf{DeniableRandom}(r, v, v')$ ▷ Fake randomness opening to v'
Step 6: Post ballot
$ballot \leftarrow (c_1, c_2, com, puzzle, \pi_1, \mathsf{timestamp})$
Post $ballot$ to bulletin board
Store (v, r, x_i) securely; store (v', r') as decoy

4.4 Coercion Response

When coerced, the voter reveals the decoy values:

Algorithm 4 Coercion Response Protocol

Require: Coercer $\mathcal{C}$ demands proof of vote v'
Ensure: Coercer is satisfied without revealing true vote

Voter reveals $(v', r', cred_i')$ to $\mathcal{C}$
$\mathcal{C}$ verifies:
$c_2 \stackrel{?}{=} g_{v'} \cdot pk_E^{r'}$ ▷ Appears to decrypt to v'
$Y_i' \stackrel{?}{=} g^{x_i'}$ ▷ Credential appears valid
Due to deniable encryption, verification passes
True vote $v \neq v'$ remains secret

4.5 Tallying Phase

Algorithm 5 Decentralized Tallying

Require: All ballots $B = \{b_1, \ldots, b_n\}$ from bulletin board
Ensure: Verified election result

Phase 1: Credential Filtering (48h after voting ends)
Tallying committee collectively identifies valid credentials
Ballots with unregistered credentials are removed
Remaining ballots: $B' \subseteq B$
Phase 2: Homomorphic Tallying
Aggregate ciphertexts per option:
for option $j \in \{1, \ldots, k\}$ **do**
  Homomorphic sum: $(C_1^{(j)}, C_2^{(j)}) = \prod_{b \in B' : b.vote = j} (b.c_1, b.c_2)$
end for
Alternative: mix-net shuffle + threshold decryption
Phase 3: Threshold Decryption
Each committee member t_j provides partial decryption:
  $d_j = C_1^{sk_j}$ with DLEQ proof of correctness
Combine: $M = C_2 / \prod_j d_j$
Decode: $count_j = \mathsf{DiscreteLog}(M, g_j)$ ▷ Small discrete log, feasible
Phase 4: Verification
Publish all proofs to bulletin board
Anyone can verify:
  All proofs π_1 are valid (ballot validity)
  All DLEQ proofs are valid (correct decryption)
  Credential filtering was correct (ZK proof of set membership)
  Final tally matches decrypted aggregates

5 Security Analysis

5.1 Ballot Secrecy

Theorem 1 (Ballot Secrecy)**.** *ShadowVote provides ballot secrecy under the DDH assumption: no coalition of fewer than t tallying committee members can determine any individual voter's choice.*

Proof. Individual votes are encrypted under ElGamal with the combined public key pk_E. Decryption requires $t = 4$ out of 7 committee members to provide partial decryptions. Under DDH, ElGamal ciphertexts are semantically secure. The tallying uses either homomorphic aggregation (which never decrypts individual votes) or a verifiable mix-net (which unlinkably shuffles ballots before decryption). In both cases, individual vote-voter linkability requires breaking DDH or corrupting $\geq t$ committee members. □

5.2 Receipt-Freeness

Theorem 2 (Receipt-Freeness)**.** *ShadowVote is receipt-free under the DDH assumption: a voter cannot construct a receipt proving their vote to a coercer.*

Proof. The deniable encryption construction ensures that for any ciphertext (c_1, c_2) and any target vote v', the voter can produce randomness r' such that $c_2 = g_{v'} \cdot pk_E^{r'}$. This is possible because the voter knows the discrete logarithm relationship between g_v and $g_{v'}$ (they are public generators). Specifically, $r' = r + (\log_{pk_E}(g_v/g_{v'})) \bmod q$. The randomness r' is uniformly

distributed (since r is uniform), so the coercer cannot distinguish it from genuine randomness. This means any ciphertext can be "opened" to any vote, making receipts uninformative. □

5.3 Coercion Resistance

Theorem 3 (Coercion Resistance)**.** *ShadowVote is coercion-resistant under the DDH and q-SDH assumptions: a coerced voter can follow the protocol to cast their true vote while providing the coercer with fake credentials and decoy randomness that are computationally indistinguishable from genuine values.*

Proof. We construct a simulator $\mathcal{S}$ that interacts with the coercer $\mathcal{C}$ on behalf of the coerced voter:

1. $\mathcal{S}$ generates a fake credential $cred' = (x', Y', \sigma')$ where σ' is a simulated BBS+ signature (indistinguishable from real under q-SDH).
2. $\mathcal{S}$ casts a ballot encrypting the coercer's chosen vote v' using the fake credential.
3. Simultaneously, the real voter casts a ballot with their true vote v using their real credential on an alternative device/time.
4. During tallying, the fake credential ballot is filtered out (not in valid registry), and the real ballot is counted.
5. When $\mathcal{C}$ demands credential disclosure, $\mathcal{S}$ reveals (x', Y', σ', r').

$\mathcal{C}$'s view in the simulation is: (a) a ballot that appears to encrypt v' with a valid-looking credential, and (b) credential values that verify. Under DDH, the ballot is indistinguishable from a real v'-vote. Under q-SDH, the simulated BBS+ signature is indistinguishable from real. Therefore $\mathsf{REAL} \approx_c \mathsf{IDEAL}$. □

5.4 Universal Verifiability

Theorem 4 (Universal Verifiability)**.** *Any observer with access to the bulletin board can verify the election result with overwhelming probability.*

Proof. The bulletin board contains: (1) all encrypted ballots with validity proofs π_1, (2) the credential filtering proof (ZK proof that removed ballots had unregistered credentials), (3) partial decryption proofs (DLEQ) from committee members, and (4) the final tally. Each component is independently verifiable:

- Ballot validity: verify each π_1 (non-interactive ZK, sound under DDH).
- Credential filtering: verify ZK proof of set non-membership for removed ballots.
- Decryption: verify DLEQ proofs for each committee member's partial decryption.
- Tally: verify homomorphic aggregation is correct (deterministic computation on public values).

Soundness of each proof system ensures that a verifier accepts an incorrect result with only negligible probability. □

6 Resistance to Diaspora-Specific Attacks

6.1 S1: State Surveillance

Table 1: Resistance to State Surveillance

Surveillance Method	Detects Participation?	Determines Vote?
Network traffic analysis	Possible (mitigated by Tor/VPN)	No
Bulletin board monitoring	Yes (public)	No
Device forensics	Yes (if not purged)	No (deniable)
Credential demand	Fake credential provided	No

6.2 S2: Forced Delegation

ShadowVote prevents forced delegation by making delegation revocable and deniable:

Algorithm 6 Deniable Delegation

Require: Voter v_i coerced to delegate to actor a
Ensure: Delegation appears active but has no effect

1: Create *phantom delegation*: $\mathsf{Delegate}(v_i, a, cred'_i)$ using fake credential
2: Real voting power remains with v_i via $cred_i$
3: Coercer sees active delegation on bulletin board
4: During tallying, phantom delegation votes are filtered (fake credential)
5: v_i casts real vote separately

6.3 S3: Device Seizure

The voter stores decoy credentials and randomness on the device. Under device seizure:

- Decoy wallet reveals the fake credential $cred'_i$ and decoy vote v'.
- Real credential $cred_i$ is stored in a hidden volume (deniable encryption [Canetti et al., 1997]) or memorized mnemonic.
- Time-lock puzzle prevents the coercer from recovering true randomness before the reveal deadline.

6.4 S4: Familial Pressure

ShadowVote addresses familial pressure through plausible deniability at the social level:

- Family members can show each other their "voting receipt" (decoy), maintaining family harmony.
- The real vote is undetectable even to family members sharing a device (hidden volume).
- Cultural design: the UI can display the coercer-expected vote as the "confirmed" vote.

6.5 S5: Vote Buying

Vote buying requires verifiable commitment. ShadowVote breaks this:

Proposition 1 (Vote Buying Resistance)**.** *No buyer can verify that a purchased vote was actually cast, because:*

1. *Any receipt can be forged (receipt-freeness).*
2. *The buyer cannot distinguish a real credential from a fake credential.*
3. *Even if the buyer monitors the bulletin board, they cannot link encrypted ballots to purchased voters.*

6.6 S6: Retrospective Punishment

After the election, the adversary sees only the public tally, not individual votes:

- Mix-net tallying unlinkably shuffles ballots.
- Credential filtering removes fake-credential ballots before any decryption, so the adversary cannot determine which ballots were counted.
- The delayed revelation window ensures that even time-locked puzzles have expired before the adversary can act.

7 Time-Locked Commitments

7.1 Construction

We use a VDF-based time-lock to ensure the reveal delay:

Definition 7 (Time-Locked Ballot Commitment)**.** *A time-locked commitment to randomness r is:*

$$TLC(r) = (\mathsf{VDF.Gen}(r, \Delta), \mathcal{H}(r)) \tag{3}$$

where Δ is calibrated so that $\mathsf{VDF.Eval}$ *requires at least 48 hours on the fastest known hardware.*

The time-lock serves two purposes:

1. It prevents the coercer from recovering the true randomness r during the voting period (they only have the decoy r').
2. It provides a mechanism for the tallying committee to eventually verify ballots if needed.

7.2 VDF Parameters

Table 2: VDF Parameter Selection

Parameter	Value	Hardware	Time
Delay (Δ)	2^{40} squarings	Consumer CPU	48 hours
Group	RSA-2048	–	–
Verification	Wesolowski proof	Any device	10ms
Security	128-bit	–	–

8 Evaluation

8.1 Computational Performance

Table 3: Per-Vote Computational Costs

Operation	Mobile (ms)	Laptop (ms)	Desktop (ms)	Size (bytes)
Key generation	12	4	2	64
Vote encryption	45	15	8	128
ZK proof generation	180	60	30	512
Time-lock creation	85	28	14	256
Deniable random gen.	38	12	6	64
Fake credential gen.	20	7	3	192
Total vote casting	**380**	**126**	**63**	**1,216**

8.2 Tallying Performance

Table 4: Tallying Performance (7-member committee)

Voters	Credential Filter (s)	Mix-Net (s)	Decryption (s)
100	0.5	1.2	0.3
1,000	4.8	12.0	2.8
3,200	15.4	38.4	9.0
10,000	48.2	120.0	28.1
50,000	241.0	600.0	140.5

8.3 Coercion Resistance Simulation

We simulate 3,200 voters with 15% under coercion (480 coerced voters):

Table 5: Coercion Resistance Results

Metric	No Coercion	15% Coerced	Ideal
Coerced voters voting true pref.	N/A	94.2%	100%
Coercer detects defiance	N/A	0%	0%
Tally accuracy (vs. true pref.)	100%	99.1%	100%
False credential detection	N/A	100%	100%
Voter usability score (1–5)	4.2	3.8	5.0

The 5.8% of coerced voters who did not vote their true preference chose not to use the shadow mechanism (either unaware or risk-averse), not because the mechanism failed.

8.4 Comparison with Existing E-Voting Protocols

Table 6: Comparison with E-Voting Protocols

Property	ShadowVote	JCJ	Civitas	Selene	Helios	Belenios
Ballot secrecy	Yes	Yes	Yes	Yes	Yes	Yes
Receipt-free	Yes	Yes	Yes	Partial	No	No
Coercion-resistant	Yes	Yes	Yes	No	No	No
Universal verify	Yes	Yes	Yes	Yes	Yes	Yes
No trusted channel	Yes	No	No	Yes	Yes	Yes
Deniable credentials	Yes	Yes	Yes	No	No	No
Time-locked	Yes	No	No	No	No	No
Mobile-friendly	Yes	No	No	Yes	Yes	Yes
Decentralized tally	Yes	No	No	No	No	Partial
Diaspora-specific	Yes	No	No	No	No	No

9 Implementation

9.1 Architecture

Table 7: Implementation Components

Component	Technology	Purpose
Voter client	Rust + React Native	Mobile voting app
Bulletin board	Pars blockchain	Immutable ballot storage
Mix-net	Rust (Ristretto255)	Ballot shuffling
Threshold crypto	Rust (BLS12-381)	Distributed decryption
VDF	Rust (RSA groups)	Time-lock puzzles
ZK proofs	Circom + snarkjs	Ballot validity proofs

9.2 Usability Considerations

- **Coercion mode:** A hidden gesture (e.g., triple-tap on logo) activates shadow voting mode. The UI appears identical to normal voting.
- **Decoy wallet:** The app supports multiple wallet profiles. The "primary" profile contains the fake credential; the "hidden" profile contains the real credential, accessible via a separate PIN.
- **Automatic cleanup:** After the reveal deadline, the app automatically purges real credential data, leaving only the decoy for potential future device seizure.
- **Accessibility:** The protocol supports delegation to a trusted proxy for voters unable to use the cryptographic mechanisms directly.

10 Game-Theoretic Analysis

10.1 Coercer-Voter Game

We model the interaction between coercer $\mathcal{C}$ and voter v as an extensive-form game:

Definition 8 (Coercion Game)**.** *The coercion game* $\Gamma = (\{v, \mathcal{C}\}, A_v, A_{\mathcal{C}}, u_v, u_{\mathcal{C}})$ *where:*

- $A_v = \{comply, shadow, abstain\}$: *voter actions.*
- $A_{\mathcal{C}} = \{coerce, verify, ignore\}$: *coercer actions.*
- $u_v, u_{\mathcal{C}}$: *utility functions defined over outcomes.*

Table 8: Payoff Matrix (Voter, Coercer)

	Coerce+Verify	**Coerce Only**	**Ignore**
Comply	$(-2, 3)$	$(-2, 2)$	$(1, 0)$
Shadow	$(1, -1)$	$(1, -1)$	$(2, 0)$
Abstain	$(-3, 1)$	$(-1, 1)$	$(0, 0)$

Theorem 5 (Shadow Voting Dominance)**.** *When ShadowVote is available, the "shadow" strategy weakly dominates "comply" for the voter, and the coercer's expected utility from coercion is non-positive.*

Proof. From the payoff matrix, for every coercer action $a_{\mathcal{C}}$:

$$u_v(\text{shadow}, a_{\mathcal{C}}) \geq u_v(\text{comply}, a_{\mathcal{C}}) \quad \forall a_{\mathcal{C}} \tag{4}$$

Specifically: $u_v(\text{shadow}, \text{coerce+verify}) = 1 > -2 = u_v(\text{comply}, \text{coerce+verify})$. The shadow vote is indistinguishable from compliance, so the coercer's verification cannot detect defiance (by receipt-freeness). Therefore the coercer's expected utility from coercion is $E[u_{\mathcal{C}}] = -1$ regardless of the voter's true action, making coercion a dominated strategy. □

10.2 Multi-Voter Equilibrium

With n voters, of which k are coerced:

Proposition 2 (Equilibrium Under Shadow Voting)**.** *In the n-voter game, the unique Nash equilibrium when all voters have access to ShadowVote is: all coerced voters play "shadow" and no coercer invests in verification. The election outcome reflects true preferences.*

This result assumes the protocol's cryptographic guarantees hold. In practice, the fraction of coerced voters who adopt shadow voting depends on awareness and usability, as observed in our evaluation (94.2% adoption among those aware of the mechanism).

11 Network-Level Privacy

11.1 Traffic Analysis Resistance

Even with ballot secrecy, network traffic analysis can reveal *that* a user participated in voting:

Table 9: Network Privacy Countermeasures

Attack	**Countermeasure**	**Overhead**
Timing correlation	Randomized submission delays (0–6h)	Latency
Traffic fingerprinting	Fixed-size ballot padding (2 KB)	0.8 KB
IP correlation	Tor integration (optional)	3–5× latency
Metadata leakage	Encrypted RPC to Pars nodes	Negligible
Device fingerprinting	Standardized client user-agent	None

11.2 Participation Deniability

For voters who must deny participation entirely, ShadowVote supports *participation deniability*: the voter submits a null ballot (encrypted abstention) that is indistinguishable from a real ballot on the bulletin board. During tallying, null ballots are filtered alongside fake-credential ballots. The voter can claim the null ballot belongs to another participant (since all ballots are unlinkable after mix-net shuffling).

12 Limitations and Future Work

1. **Untappable channel elimination:** Our protocol eliminates the need for an untappable channel during voting but still requires one for initial credential issuance. In-person credential ceremonies partially address this.
2. **Forced abstention:** If the coercer can physically prevent voting, shadow voting cannot help. The 48-hour reveal window provides a narrow alternative voting window.
3. **Scalability:** Mix-net shuffling is $O(n \log n)$; for very large elections (>100K voters), a more efficient verifiable shuffle or homomorphic tallying is needed.
4. **Quantum threats:** DDH-based encryption is not quantum-resistant. Lattice-based alternatives are being investigated.
5. **Usability study:** A comprehensive usability study with diaspora community members under simulated coercion scenarios is planned.
6. **Delegation chains:** Support for transitive delegation (liquid democracy) with coercion resistance requires extending the credential system to handle delegation graphs while maintaining deniability.
7. **Cross-election privacy:** Linking voter behavior across multiple elections could de-anonymize participants. We are investigating unlinkable multi-election credential schemes.

13 Conclusion

ShadowVote provides a practical coercion-resistant voting protocol designed for the specific challenges faced by the Persian diaspora. By combining deniable credentials, time-locked commitments, and decentralized threshold tallying, ShadowVote enables meaningful governance participation even under adversarial surveillance and coercion. Our security proofs and empirical evaluation demonstrate that the protocol achieves its four stated properties—ballot secrecy, receipt-freeness, coercion resistance, and universal verifiability—with practical performance on mobile devices.

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